Cranberry and blueberry: evidence for protective effects against cancer and vascular diseases.

Growing evidence from tissue culture, animal, and clinical models suggests that the flavonoid-rich fruits of the North American cranberry and blueberry (Vaccinium spp.) have the potential ability to limit the development and severity of certain cancers and vascular diseases including atherosclerosis, ischemic stroke, and neurodegenerative diseases of aging. The fruits contain a variety of phytochemicals that could contribute to these protective effects, including flavonoids such as anthocyanins, flavonols, and proanthocyanidins; substituted cinnamic acids and stilbenes; and triterpenoids such as ursolic acid and its esters. Cranberry and blueberry constituents are likely to act by mechanisms that counteract oxidative stress, decrease inflammation, and modulate macromolecular interactions and expression of genes associated with disease processes. The evidence suggests a potential role for dietary cranberry and blueberry in the prevention of cancer and vascular diseases, justifying further research to determine how the bioavailability and metabolism of berry phytonutrients influence their activity in vivo.